



Abound Edge MQTT Interface for data streaming

Version : 0.8_WOA

Date : 27/03/2025

Disclaimer: Please note that the Abound Insights platform, formerly known as the CORTIX platform, is undergoing changes to deliver a better experience to you. You may come across mentions of the CORTIX platform in our documents, videos and other material. We appreciate your understanding and patience as we complete the changes.

Version	Date	Author	Description
0.1	12/09/2021	Dattaguru B.N	Initial draft
0.2	15/10/2021	Dattaguru B.N	Added Equipment Data format
0.3	19/08/2022	Dattaguru B.N	Updated for subscription section for customer to get the data from Abound Edge MQTT broker
0.4	23/08/2023	Dattaguru B N	Updated with openmqtt format for customers to publish into Abound Edge MQTT broker
0.5	07/09/2023	Dattaguru B N	Updated with Metadata details and “info” section removed from data payload
0.6	25/09/2023	Dattaguru B N	Topic standardization for all kinds of message types
0.7	18/01/2024	Dattaguru B N	Updated Metadata section with additional fields. Topic format changed to indicate data flow direction
0.8	13/12/2024	Dattaguru B N	Updated Metadata section with enhanced Metadata format. Abound re-branding.
0.8_WOA	27/03/2025	Dattaguru B N	Updated Metadata section with “org” object, re arrangement of few fields and standardizing mandatory keys. Added hyperlink for example and added schema that helps developers/integrators.

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About Abound Edge MQTT Broker

The purpose of this document is to provide high level specification for the Abound Edge MQTT Broker specification for data streaming from and to third party systems. We expect the readers of this document will already have an understanding of MQTT concepts.

The Abound Edge MQTT Broker facilitates and acts as bridge between data generators and the data consumers that churn the data to create insights. Data generators may include edge controllers, BMS controllers or cloud applications that are capable of publishing to MQTT brokers. Data consumers may include any such software service that consumes data by subscribing to the MQTT brokers.

Abound Edge MQTT broker is part of the Abound Edge software product which enables plug-and-play connectivity of equipment and building management systems. It includes edge and cloud applications to enable bi-directional data exchange and data transformation to Abound Insights and other platforms. It includes device management functionality for diverse types of edge agents connected.

The Abound Insights Platform is an AI and IoT platform for predictive maintenance and energy efficiency. It simplifies equipment maintenance and energy efficiency activities by providing an AI-powered assistant. The assistant helps prioritize actions, offers prescriptive recommendations, and simplifies workflows for facility managers and technicians.

Third party systems can subscribe to the broker to receive transformed equipment data or publish their equipment data into broker. We recommend 15 min publishing interval for clients that are publishing to the broker. For the subscriber clients, Abound Edge services publish transformed equipment data that is created from multiple connected devices and transformed to logical equipment.

MQTT Data Interface

Abound Edge data services support MQTT interfaces for data exchange between two parties by exchanging the data through Abound Edge MQTT Broker datan.cortixedge.io.

Data type of the points shall be signed off in the initial phase of the project as all data received and sent are in String data type. Hence, conversions shall be planned accordingly by respective clients.

It currently listens on the following ports. User credentials will be shared during implementation.

5083 : MQTT over encrypted SSL.

Security at Abound Edge MQTT broker

Abound Edge MQTT broker

- hosted behind the firewall and ensures avoiding direct access to broker.
- hosted over SSL with TLS encryption.
- can connect only to authorized clients that have valid credentials.
- topics can be accessed to publish only by authorized clients.
- topics can be accessed to subscribe only by authorized clients.
- can be connected by clients using client certificates (optional)

MQTT Publish

Publish is a process in which device / client sends data to the broker in format explained in Exchange section 0

To publish data, client needs to follow below steps.

- Create a MQTT client with unique client ID. Please note, if same client ID is used for two different connections, Broker will disconnect the other one.
- Set the QoS level.
- Set the topic to which data need to be sent.
- Provide user credentials*
- Set the publishing interval.
- Schedule MQTT client to send the data at publishing interval.

* User credentials will be shared during implementation

MQTT Subscribe

Subscribe is a process in which device / client listens to the broker and consumes received data from the broker on a defined topic. The format is explained in Exchange section 0.

To subscribe, client needs to follow below steps:

- Create a MQTT client with unique client ID. Please note, if same client ID is used for two different connections, Broker will disconnect the other one.
- Set the topic from where data need to be received.
- Provide user credentials*
- Set subscribe time out period.
- Subscribe MQTT client to receive the data as soon as publisher publishes the data to broker.

* User credentials will be shared during implementation

Topics

MQTT topic is a string used in the MQTT protocol to identify and route messages. It is a key element in communication between MQTT publishers and subscribers to exchange messages. Abound Edge follows below standardization in defining the topic name

<message_type>/<asset_group>/<data_direction>/<controller_id>/<payload_type>

Topic fields	Description
message_type	Field for type of messages being exchanged. This can take CORTIXedgeData, CORTIXedgeMeta and CORTIXedgeTask
asset_group	Unique identifier for the account/customer/data source with which messages are being exchanged. We shall provision this value at the initiation of the project.
data_direction	This field can take either “i” or “o.” i – Indicates exchange of data is inwards to Abound Edge environment o – Indicates exchange of data is Outwards from Abound Edge environment
controller_id	Unique identifier (preferably serial number) for the source of the data like aggregators, controllers or any IOT enabled equipment that can communicate to cloud MQTT
payload_type	Optional. Field to identify the type of payload that is being exchanged. Some of the example values for this field are “openmqtt”, “signaling”, “transformed”, etc.

Exchange

Exchange is a handshake of messages between two clients (publisher and subscriber) via Topics defined on MQTT Broker.

Type of messages

- Data
- Information
- Instruction

Data exchange

Data exchange refers to the exchange of information that contains parameters along with time stamp and their values.

These types of data are exchanged at a higher frequency. Recommended frequency is 15 min.

This kind of exchange is identified by keyword “CORTIXedgeData” in the topic as below
CORTIXedgeData/<asset_group>/i/<controller_id>/<payload_type>

Device/equipment data

Device/Equipment data refers to the data generated at the source without any transformation. In the MQTT topic, this kind of data is identified with the keyword “*openmqtt*” and should be used for data Ingestion only and hence these topics will always have direction key word as “i”.

Topic

CORTIXedgeData/<account>/i/<controller_id>/openmqtt

Topic example

Consider an account/customer with name “**ABC**” having a device “**80284455**” that is sending data without transformation is represented as below

CORTIXedgeData/ABC/i/80284455/openmqtt

Typical Use case

This topic and format shall be used by any client who intends to share/publish the data to Abound Edge MQTT broker using credentials provided.

Payload format

Abound Edge MQTT broker defines below JSON format for customer to publish the payload of individual controllers:

```
{
  "livedata": {
    "_id": "XYZ9999999",
    "_unixtime": "1672895853000",
    "devices": [
      {
        "_comm": "0",
        "_id": "ABC8888888",
        "data": {
          "indoor temp": "73.5",
          "Heating setpoint": "68",
          "Cooling setpoint": "74",
          "Heating stage01 command": "0",
          "Heating stage02 command": "0",
          "Cooling stage01 command": "0",
          "Cooling stage01 command": "0",
        }
      }
    ],
    "_comm": "0",
    "_id": "ABC8888889",
    "data": {
      "indoor temp": "73.5",
      "Heating setpoint": "68",
      "Cooling setpoint": "74",
      "Heating stage01 command": "0",
      "Heating stage02 command": "0",
      "Cooling stage01 command": "0",
      "Cooling stage01 command": "0",
    }
  }
}
```

JSON Object	Mandatory/Optional	Description
Livedata	Mandatory	Mandatory field. Keyword for the parent JSON object which identifies data being shared is the live data
_id	Mandatory	Mandatory field. Keyword for unique identifier of corresponding JSON Object. Example: "_id" under "livedata" for the Controller (Can be Building Id also in cases like Commercial buildings) and "_id" under "devices" for specific device connected to controller. Use the same "_id" if both are same.
_unixtime	Mandatory	Mandatory field. Keyword for time stamp of corresponding JSON Object. Example: "_unixtime" under "livedata" for the Controller/aggregator which aggregates data from all connected devices and adds this time stamp before publishing
Devices	Mandatory	Mandatory field. Keyword for Array of JSON Objects, each object representing individual devices connected to controller.
_comm	Mandatory	Mandatory field. Keyword for communication of corresponding JSON Object. Example: "_comm" under "livedata" for the Controller and "_comm" under "devices" for specific device connected to controller 0 – represents communicating 1 – represents non communication / unavailable If no data is available, "1" should be sent for this field
Data		Mandatory field. Keyword for array of data from connected

Transformed data

Transformed data refers to the format where data generated at the source are transformed and customized to make it more informative. In the MQTT topic, this kind of data is identified with the keyword “*transformed*” and is intended for data receipt from Abound Edge MQTT broker and hence these topics will always have direction key word as “o”.

Topic

CORTIXedgeData/<account>/o/<controller_id>/transformed

Topic example

Consider an account/customer with name “**ABC**” having a device “**80284455**” that is sending data which gets transformed based on configuration and customization is represented as below

CORTIXedgeData/ABC/o/80284455/transformed

Typical Use case

This topic and format shall be used by any client who intends to receive/subscribe the transformed data from Abound Edge MQTT broker using credentials provided.

Payload format

Payload is generated in Equipment data format by converting connected devices data to logical equipment of interest.

We recommend customer to use this format to receive the data as this will be more concise and customizable with respect to data points.

```
{
  "_id": "XYZ9999999",
  "_ts": "1677585600183",
  "data": [
    {
      "_id": " ABC8888888",
      "_ts": "1677585600183",
      "_comm": "0",
      "ZNTMP": "73.5",
      "HTSP": "68",
      "CTSP": "74",
      "CMP1STS": "1",
      "CMP2STS": "0",
      "HTS1STS": "0",
      "HTS2STS": "0"
    },
    {
      "_id": " ABC8888889",
      "_ts": "1677585600183",
      "_comm": "0",
      "ZNTMP": "73.5",
      "HTSP": "68",
      "CTSP": "74",
      "CMP1STS": "1",
      "CMP2STS": "0",
      "HTS1STS": "0",
      "HTS2STS": "0"
    }
  ],
  "info": {
    "ServArea": "Sales",
    "cntlNrNw": "BacNet",
  }
}
```

JSON Object	Description
_id	Mandatory field. Keyword for unique identifier of corresponding JSON Object.
_ts	Mandatory field. Keyword for time stamp of corresponding JSON Object. Time stamp is represented in UTC time format (Unix time)
_comm	Mandatory field. Keyword for communication of corresponding JSON Object. Value sent is determined by availability of data. 0 – represents communicating 1 – represents non communication / unavailable If no data is available, “1” would be sent for this field
data	Mandatory field. Keyword for JSON Array that contains transformed data with corresponding values.
Info	Optional field. Any additional information that are specific to consumer. This field is normally empty and is not preferred.

Information exchange

Metadata is an optional and additional but may include datapoints, their data types, make and model of devices/ equipment, location important information that are rarely changed. This information, etc.

Meta data is exchanged between two clients (publisher and subscriber) via Topics defined on MQTT Broker. Metadata should be shared across only on changes or at very low frequency.

We recommend once in a day.

This kind of exchange is identified by keyword “CORTIXedgeMeta” in the topic as below

Topic

CORTIXedgeMeta/<account>/i/<controller_id>/<payload_type>

Topic example

Consider an account/customer with name “**ABC**” having a device “**80284455**” that is sending meta data in payload type (optionally v0.8_WOA) information into Abound Edge MQTT broker is represented as below

CORTIXedgeMeta/**ABC**/i/**80284455**

If metadata is being sent out from Abound Edge MQTT broker, then topic will be as below

CORTIXedgeMeta/**ABC**/o/**80284455**

Payload format

MetaData payload is the payload that will have information about the data points such as its datatype, device/equipment make and models, software versions, etc. This payload should be published only on change and should not be exchanged at data frequency as these values seldom change.

Abound Edge MQTT broker defines below JSON format for customer to publish the payload of individual controllers:

In the payload, keywords that start with “_” are mandatory. Unless otherwise specified, all values are sent as Strings (Double quoted)

```
{
  "$schema": "http://json-schema.org/draft-04/schema",
  "type": "object",
  "properties": {
    "_ver": {"type": "string"},
    "_conID": {"type": "string"},
    "_acclID": {"type": "string"},
    "_subID": {"type": "string"},
    "_org": {"type": "object",
      "properties": {
        "_id": {"type": "string"},
        "_name": {"type": "string"},
        "_contact": {"type": "string"},
        "_email": {"type": "string"},
        "info": {"type": "object"}
      }
    },
    "required": [
      "_id",
      "_name",
      "_contact",
      "_email",
      "info"
    ],
    "_locations": {"type": "array",
      "items": [{"type": "object",
        "properties": {
          "_id": {"type": "string"},
          "_siteName": {"type": "string"},
          "_address": {"type": "string"},
          "_latitude": {"type": "string"},
          "_longitude": {"type": "string"},
          "_tempUOM": {"type": "string"},
          "_timeZone": {"type": "string"},
          "info": {"type": "object"},
          "_assetList": {"type": "array",
            "items": [{"type": "object",
              "properties": {
                "_id": {"type": "string"},
                "_type": {"type": "string"},
                "info": {"type": "object"}
              },
              "required": [
                "_id",
                "_type",
                "info"
              ]
            }
          ]
        },
        "required": [
          "_id",
          "_siteName",
          "_address",
          "_latitude",
          "_longitude",
          "_tempUOM",
          "_timeZone",
          "info",
          "_assetList"
        ]
      }
    ]
  },
  "required": [
    "_ver",
    "_conID",
    "_acclID",
    "_subID",
    "_org",
    "_locations",
    "_assetList",
    "info"
  ]
}
```

Key	Description
_ver	Indicates format version.
_conID	Mandatory field. Unique identifier for the connection. (to be provided by Carrier ABOUND). Can be individual connection Ids for individual customers or One connection id for logical group within a customer.
_acclID	Mandatory field. Unique customer group identifier maintained in ABOUND (to be provided by Carrier ABOUND). Customer group can be configured based on dealer or any other logical grouping.
_subID	Mandatory field. Unique customer group subscription identifier maintained in ABOUND. (to be provided by Carrier ABOUND). One customer group can have one or many subscriptions and one subscription can have one or many sites.
_org	Mandatory field. Key to identify Organization details. <i><u>_id</u>: Company ID, Unique customer identifier maintained in other systems. Should represent actual end customer with a list of sites and equipment.</i> <i><u>_name</u>: Company Name</i> <i><u>_contact</u>: Company contact person name</i> <i><u>_email</u>: Company contact person email id</i>
_locations	Mandatory field. Array of locations and its details where thermostat/controller is located <i><u>_siteID</u>: Friendly site identifier used by customer</i> <i><u>_siteName</u>: Friendly name used by customer</i> <i><u>_address</u>: Should be less than 50 characters</i> <i><u>_latitude</u>: Latitude of the location</i> <i><u>_longitude</u>: Longitude of the location</i> <i><u>_timezone</u>: https://timezonedb.com/time-zones</i>
_assetList	Mandatory field. Keyword for array of assets/controllers inside a location and connected to Integration service <i><u>_id</u>: Unique identifier for connected asset/controller (usually Serial number)</i> <i><u>_type</u>: Type of asset supported by ABOUND Insights (to be provided by Carrier ABOUND). For available types, please refer to Appendix for sample JSON data and the applicable asset list types shared along with this document.</i>
info	Keyword for any additional information under each of the objects. This provides more detail about respective object and helps better information about the same

For sample JSON with example data, please refer to **Appendix for sample JSON data and the applicable asset list types** shared along with this document.

NOTE: All the keys are mandatory. Keys that start with “_” cannot be empty.

Instruction exchange

Instruction messages contain instructions passed to the receiver to perform certain set of activities or tasks. This may include instructing a thermostat to change the setpoint, restarting of the connected equipment, etc.

Instructions are exchanged between two clients (publisher and subscriber) via Topics defined on MQTT Broker.

Instruction should be exchanged at very low frequency and is only on demand basis. This kind of exchange is identified by keyword “CORTIXedgeTask” in the topic as below

Topic

CORTIXedgeTask/<account>/o/<controller_id>/<payload_type>

Topic example

Consider an account/customer with name “**ABC**” having a device “**80284455**” that is instructed to send “onDemand” data to Abound Edge MQTT broker is represented as below

CORTIXedgeTask/ABC/o/80284455/onDemand

Payload format

Payload varies based on the instruction or tasks being assigned to the client. This is currently used only by SmartConnects developed by Abound Insights and is not available for public.

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